

LeakSignal

Final Detection Logic Architecture — Summary & Comparison

Final Status: The detection-logic architecture is ready to proceed with. Exact thresholds, time windows, lookback periods, transfer definitions, and risk-score weights can be tuned during implementation.

1. Core Idea

LeakSignal does not judge a network event in isolation. It identifies the host, understands its normal behaviour, quickly evaluates the new event, gathers deeper relevant history only when needed, and correlates events by host, destination, timing, repetition, and persistence before making an explainable decision.

Final principle: “Identify the host → understand its normal behaviour → quickly evaluate the new event → gather deeper relevant history only when needed → correlate events over time → detect repetition and persistence → make an explainable decision.”

2. Final Detection Workflow

- 1. Event Validation & Normalization — validate required fields, timestamps, traffic direction, bytes, and normalize host/destination identity.
- 2. Host Identity Resolution — identify the host using a stable host identity and retain IP as network context.
- 3. Get / Create Persistent Host Profile — maintain behavioural baseline, timing behaviour, destination history, transfer behaviour, recent activity, and risk state.
- 4. Quick Context Check — check unusual timing, new/rare destination, and unusual transfer behaviour.
- 5. Initial Suspicion Decision — decide whether deeper analysis is required.
- 6. Dynamic Deep Context Gathering — if suspicious, query only relevant history for the same host, same/related destination, similar time window, and relevant lookback period.
- 7. Event Correlation — identify related repeated events.
- 8. Repetition & Persistence Analysis — determine whether the activity continues in a meaningful pattern across time.
- 9. False-Positive Check — account for legitimate scheduled backups, approved cloud services, synchronization, updates, and normal automated behaviour.
- 10. Evidence Combination — combine confirmed signals.
- 11. Exfiltration Risk Score (ERS) — generate an explainable risk assessment.
- 12. Final Decision — classify as Normal, Monitor, Suspicious, or Possible Slow Data Exfiltration.
- 13. Update Host Profile — update activity, destinations, timing behaviour, risk state, and alert history.

3. Persistent Host Context

Each host has a separate logical profile in the database rather than a separate physical file. The profile provides organized, persistent context while keeping all events linked through a stable host identifier.

- Stable host identity
- Behavioural baseline
- Normal timing pattern
- Destination history
- Transfer behaviour
- Recent activity summary
- Historical activity context
- Risk and alert state

4. On-Demand Deep Context Gathering

This is the major efficiency improvement. Every event receives a fast initial check, but the system does not perform expensive historical analysis for every normal event.

Example: PC-001 sends 5 MB to a new destination at 02:15 AM. The transfer may be individually harmless, but the destination is new and the time is unusual. LeakSignal then queries only relevant history: the same host, destination, similar time window, and recent days. If similar events appear repeatedly across multiple days, the system can identify a persistent slow-drip pattern.

Efficiency principle: Persistent summary context + targeted historical queries.

5. First Architecture vs Final Architecture

Area	First Architecture	Final Architecture
Host context	Gathered during analysis	Persistently maintained
Historical access	Potentially frequent	Deep queries only when needed
Initial processing	Direct analysis	Fast quick-context check
Live monitoring	Possible	Optimized for continuous monitoring
Host memory	General historical lookup	Persistent host-specific profile
Correlation	General event correlation	Targeted host + destination + timing correlation
Repetition	Detection signal	Dedicated analysis stage
Persistence	General concept	Dedicated analysis stage
False positives	Less explicitly structured	Dedicated verification stage
Decision process	Risk score / alert	Multi-stage evidence-based decision

6. Major Change

The final architecture does not change the original goal of LeakSignal. The major change is **when and how context is accessed**.

First architecture: Every event can gather context and then be analysed.

Final architecture: Every event first uses fast persistent host context; only suspicious or developing patterns trigger deeper, targeted historical investigation.

In one sentence: The first architecture is more query-first, while the final architecture is context-first with query-on-demand.

7. Why the Final Architecture Is Better

- **More efficient for live monitoring:** normal events are processed quickly and deeper queries are reserved for suspicious activity.
- **Host-specific behaviour:** every host develops its own baseline instead of relying on one global definition of normal.
- **Better slow-drip detection:** individually harmless events can be connected using host, destination, timing, repetition, and persistence.
- **More efficient database usage:** targeted queries retrieve only relevant historical events instead of repeatedly scanning broad history.
- **Better scalability:** persistent summaries support fast initial checks while deeper analysis remains focused.
- **Better manual investigation:** an analyst can inspect a host's behaviour, destinations, timing, activity history, and risk state in one logical profile.
- **Better false-positive control:** legitimate scheduled or automated behaviour is checked before the final classification.
- **More explainable decisions:** the final risk score is based on combined evidence rather than one isolated anomaly.

8. Final Verdict

Recommended architecture: Persistent Host Profile + Quick Initial Analysis + On-Demand Deep Context Gathering + Targeted Event Correlation.

The first architecture was a strong conceptual detection workflow, but the final architecture is more structured and practical for LeakSignal because it adds persistent host memory, efficient live processing, targeted historical investigation, dedicated repetition and persistence analysis, and explicit false-positive handling.

Final rating: The first architecture is a good basic foundation; the final architecture is the better choice for LeakSignal's intended live, long-term, slow data-exfiltration monitoring.

Final Detection Principle: LeakSignal continuously maintains host-specific behavioural context, performs a fast initial evaluation for every incoming event, dynamically gathers only relevant historical context when suspicion exists, and correlates host, destination, timing, repetition, and persistence to identify possible slow data exfiltration before generating an explainable risk decision.

Conclusion: Yes — this architecture is good to go for the detection-logic phase. The remaining work is implementation-level tuning of thresholds and parameters, not a redesign of the architecture.